

Keywords: robotic manipulation • sim-to-real transfer • synthetic data • generalization benchmark

RoboSynChallenge: Mastering Real-World Dexterity via Generalizing Synthesized Manipulation Skills

NeurIPS 2026 Competition Pairs Synthetic-Only Training with Unseen Real-World Evaluation to Benchmark Generalizable Robot Policies at Scale

By Runyi Zhao, Ruixin Wu, Chengkun Li, Hongrui Zhang, Ang Li, Ruixing Jin, Yueci Deng, Yingying Guo, et al.

arXiv:2608.12416 • NeurIPS 2026 Competition Track

Generalizable robotic manipulation remains blocked by the scarcity and narrow diversity of real-world demonstration data. RoboSynChallenge (NeurIPS 2026 Competition Track) introduces a unified benchmark pairing large-scale synthetic data generation with standardized real-world evaluation on unseen environments: participants train policies from synthesized state-action trials, and final assessment occurs exclusively on real manipulation tasks they have never seen.

AT A GLANCE

Problem: Generalizable robot manipulation policies are bottlenecked by scarce, narrow real-world data; synthetic data is abundant but sim-to-real transfer is unreliable.

Why it matters: Standardized evaluation of sim-to-real generalization at scale does not exist; RoboSynChallenge provides it for the first time at NeurIPS scope.

Method: Participants train on a large synthetic state-action corpus; evaluation is conducted exclusively on unseen real-world manipulation environments.

Result: The competition provides a three-tier difficulty benchmark (basic, dexterous, open-vocabulary) and cross-institutional comparison of policy approaches.

Evidence: NeurIPS 2026 Competition Track participation ensures broad community coverage spanning diffusion policies, flow matching, and VLA architectures.

The Big Picture

The robotics community has long believed that data scarcity is the core obstacle to generalizable manipulation: if we had enough diverse demonstrations, robot policies would generalize the way vision models did after

ImageNet. Synthetic data is the obvious supply-side fix — simulators are cheap, parallelizable, and can generate arbitrary diversity of objects, grasp configurations, and physics parameters.

The catch is sim-to-real transfer. Policies trained on synthetic data encounter the real world and fail: the physics is subtly wrong, the visual domain is shifted, the robot’s dynamics differ from the simulated model. Decades of research have produced methods for closing the sim-to-real gap — domain randomization, system identification, residual dynamics learning — but evaluating how well any of these methods generalizes across tasks has been impossible without a standardized benchmark.

RoboSynChallenge is that benchmark.

Why Existing Approaches Fall Short

Existing robotics benchmarks suffer from one of two failure modes:

Real-data benchmarks (e.g., RoboMimic, Language-Table) evaluate in-distribution generalization: test tasks are drawn from the same distribution as training demonstrations. They measure *learning efficiency*, not *generalization beyond the training distribution*.

Simulation benchmarks (e.g., RLBench, MetaWorld) remove the sim-to-real problem by evaluating inside the simulator. They measure *task variety* but not the critical question: does this work on a real robot in the real world?

RoboSynChallenge is the first benchmark to pair synthetic training with real-world evaluation at the scale of a major venue competition, providing a clean measurement of the full sim-to-real generalization pipeline.

Core Mechanism

The challenge is structured as a two-phase competition:

Phase 1 (Synthetic Training): Participants receive a large-scale corpus of (s_t, a_t) state-action pairs generated in simulation across diverse object categories, task types, and environment configurations. Tasks include:

- Pick-and-place with novel objects and arrangements
- Articulated object manipulation (cabinets, bottles, drawers)
- Tool use requiring contact-rich dexterous control

Participants may augment with additional synthetic data using provided simulators.

Phase 2 (Real-World Evaluation): Trained policies are deployed on a standardized physical robot platform. All evaluation tasks are unseen — new objects, new goal configurations, new environments. Evaluation is fully standardized across participants: identical robot hardware, identical evaluation protocol, identical success metrics.

Technical Formulation

A manipulation policy $\pi : \mathcal{O} \times \mathcal{G} \rightarrow \mathcal{A}$ maps observations $o \in \mathcal{O}$ and goal specifications $g \in \mathcal{G}$ to actions $a \in \mathcal{A}$. The challenge tests π on the distribution shift:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{(o,g) \sim \mathcal{D}_{\text{real}}} [\text{Succ}(\pi, o, g)]$$

where $\mathcal{D}_{\text{real}}$ is the real-world evaluation distribution, but training data comes exclusively from $\mathcal{D}_{\text{syn}}$ (the synthetic corpus).

The generalization gap is defined as:

$$\Delta_{\text{gen}} = \mathbb{E}_{\mathcal{D}_{\text{syn}}} [\text{Succ}] - \mathbb{E}_{\mathcal{D}_{\text{real}}} [\text{Succ}]$$

Minimizing Δ_{gen} is the core objective.

Three evaluation tiers reflect difficulty:

Tier 1: $\mathcal{T}_1$ (basic), Tier 2: $\mathcal{T}_2$ (dexterous), Tier 3: $\mathcal{T}_3$ (open-vocabulary)

with $\text{Succ}(\mathcal{T}_3) \leq \text{Succ}(\mathcal{T}_2) \leq \text{Succ}(\mathcal{T}_1)$ for all known baselines.

Learning or Inference Procedure

The competition is open to any policy architecture and training procedure, subject to the constraint of synthetic-only training. Expected approaches include:

- **Diffusion policies:** Generate action trajectories via score matching
- **Flow matching policies:** Conditional flow matching for action prediction
- **Vision-Language-Action (VLA) models:** Combine foundation model representations with action prediction heads
- **Domain randomization:** Randomize simulator physics/appearance to cover the real-world distribution

What the Guarantee Says

The challenge provides no formal guarantee about the generalizability of any policy architecture. It provides the *measurement infrastructure* to empirically determine which synthetic data generation strategies, policy architectures, and transfer techniques generalize best to unseen real-world tasks. The benchmark acts as a scoreboard for the field’s progress on this open problem.

Experimental Findings

As a competition paper, RoboSynChallenge reports baseline results rather than final participant results:

- **Tier 1 (basic manipulation):** Strong baselines achieve >60% success rate with domain randomization; simple pick-and-place transfers well.

- **Tier 2 (dexterous tasks):** Success rates drop to 20–35% for all baselines; contact-rich manipulation is the primary failure mode.
- **Tier 3 (open-vocabulary):** Below 10% success for all baselines, establishing a challenging frontier for VLA and instruction-following approaches.
- **Generalization gap:** $\Delta_{\text{gen}} > 40\%$ for all baselines, confirming substantial sim-to-real transfer challenge.

Ablations and Interpretation

Data diversity vs. volume: Preliminary analysis suggests diversity of object categories matters more than raw data volume for Tier 2–3 transfer, consistent with findings in the vision literature.

Visual vs. proprioceptive sim-to-real: Visual domain gap accounts for ~60% of the generalization gap for Tier 1 tasks; proprioceptive and contact dynamics gaps dominate for Tier 2–3.

Action representation: Policies trained with relative action targets transfer better than absolute-coordinate policies, suggesting coordinate-space choice is a significant source of sim-to-real failure.

Reference

Runyi Zhao, Ruixin Wu, Chengkun Li, Hongrui Zhang, Ang Li, Ruixing Jin, Yueci Deng, Yingying Guo, Lihe Ding, Shaocong Dong, Tianfan Xue, Yanjun Gao, Yudong Luo, Pascal Poupart, Simo Wu, Kui Jia, Wei-shi Zheng, Guiliang Liu. **RoboSynChallenge: Mastering Real-World Dexterity via Generalizing Synthesized Manipulation Skills**. arXiv:2608.12416, August 2026. NeurIPS 2026 Competition Track. <https://arxiv.org/abs/2608.12416>